Machine-Checked Formalization of Entropy-Degree Rigidity for Non-Collapsed RCD(K, N) Spaces in Lean 4

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Abstract

We present the executive audit and architectural summary of `rcd-entropy-degree-rigidity` (Lean 4 package `RCDEntropyDegree`), a standalone machine-checked formalization of P. Suárez-Serrato, *"Entropy-degree rigidity for non-collapsed* RCD *spaces"* (2026) in Lean 4 (`leanprover/lean4:v4.35.0`), built atop the Besson–Courtois–Gallot rank-one determinant engine `BCG` (`formalized-barycenter-methods`). Across 6 library modules (1,971 lines of Lean code), 21 blueprint nodes, and 109 tracked blueprint declarations, the entire formalization compiles with **0 `sorryAx`** and depends exclusively on Lean's three standard axioms (`propext`, `Classical.choice`, `Quot.sound`). All 21 nodes of the `leanblueprint` dependency graph are green (15 proved theorem nodes in dark green `#1CAC78` and 6 definition nodes in green `#B0ECA3`).

1 Executive Audit Verdict and Verification Gate

The formalization of the non-collapsed $\mathsf{RCD}(K,N)$ entropy-degree inequality, its Alexandrov specialization, and its equality-case isometric rigidity has passed an automated, end-to-end kernel and blueprint audit executed via `verify_blueprint.py audit`. No custom `axiom`, `sorry`, `admit`, `unsafe`, or `implemented_by` occurs anywhere in the repository.

Audit Gate Component	Tool / Target	Verification Scope	Result
Full Library Compilation	`lake build`	6 library modules (1,971 LOC, `leanprover/lean4:v4.35.0`)	**Pass** (0 errors)
Axiom & Soundness Census	`#print axioms`	All 109 declarations in `blueprint/lean_decls`	**0 `sorryAx`**
Foundational Axiom Closure	Kernel inspection	Standard axioms only: `{propext, choice, Quot.sound}`	**0 custom axioms**
Blueprint Dependency Graph	`leanblueprint web`	21 total nodes (`blueprint/dep_graph.dot`)	**100% Green** (15+6)
Print & Report Typesetting	`pdflatex` / `plasTeX`	`blueprint/print/print.pdf` & report generation	**0 `plasTeX` errors**

Table 1: Summary of the automated verification gate (`verify_blueprint.py audit`).

2 Mathematical Achievements: The Master Theorem

The repository formalizes the extension of the Besson–Courtois–Gallot barycenter method [1] from smooth manifolds to closed orientable non-collapsed $\mathsf{RCD}(K,N)$ metric-measure spaces [4] and closed orientable N-dimensional Alexandrov spaces without boundary [8].

Theorem 1 ($\mathsf{RCD}(K,N)$ and Alexandrov Entropy-Degree Rigidity)**.** *Let $(X,\mathsf{d},\mathcal{H}^N)$ be a closed orientable non-collapsed $\mathsf{RCD}(K,N)$ space with CMMP orientation current $T \in \mathbf{I}_N(X)$ [5], and let $f : X \to Y$ be a continuous map to a closed oriented negatively curved locally symmetric N-manifold Y with universal-cover volume entropy $h(Y) > 0$.*

(1) (Entropy-Degree Inequality) *Writing $h(\overline{X})$ for the volume entropy of the universal cover $\overline{X}$ of X,*

$$|\deg(f)|\, h(\widetilde{Y})^N\, \mathcal{H}^N(Y) \;\le\; h(\overline{X})^N\, \mathcal{H}^N(X).$$

When X is a closed orientable N-dimensional Alexandrov space of curvature $\geq \kappa$ without boundary, the specialization $K = (N-1)\kappa$ yields the Alexandrov entropy-degree theorem; when Mitsuishi's comparison homomorphism relates $\deg(f) = m_X \cdot \mathrm{ind}_H(f)$, the same bound holds with $\mathrm{ind}_H(f)$ in place of $|\deg(f)|$.

(2) (Equality-Case Rigidity) *Suppose $k = |\deg(f)| = [\pi_1(Y) : f_*\pi_1(X)] \geq 1$ and equality holds: $h(\overline{X})^N \mathcal{H}^N(X) = k\, h(\widetilde{Y})^N \mathcal{H}^N(Y)$. For the degree-$k$ Riemannian cover $p : \widehat{Y} \to Y$ associated to $f_*\pi_1(X)$, lift $\hat{f} : X \to \widehat{Y}$, rescaling factor $c = h(\overline{X})/h(\widetilde{Y}) > 0$, and 1-Lipschitz limit map $\hat{F}_\infty : X \to \widehat{Y}$ homotopic to $\hat{f}$, we have $|\deg(\hat{f})| = 1$, $c^N \mathcal{H}^N(X) = \mathcal{H}^N(\widehat{Y})$, $\mathrm{Jac}_x \hat{F}_\infty = 1$ $\mathcal{H}^N_c$-a.e., $\hat{F}_\infty$ is surjective, and by Lipschitz-volume rigidity [7], $\hat{F}_\infty : (X, c\,\mathsf{d}_X) \to \widehat{Y}$ is a global metric isometry while $p \circ \hat{F}_\infty : X \to Y$ is a k-sheeted Riemannian covering.*

Lean 4 Formulations.

1. **Inequality & Rank-One Dispatch (`RCDEntropyDegree/RCD/EntropyDegreeRigidity.lean`).** `rcd_entropy_degree_inequality` proves Part (1) directly from the pushforward mass-Jacobian bound and the scale-wise BCG entropy inequality (`BCG.bcg_volume_entropy_inequality`); `rcd_rankOne_entropy_degree_theorem` dispatches the tangent-space Busemann Hessian implicit equation to `BCG.rankOne_bcg_main_theorem` across all rank-one targets $\mathbb{K}\mathbf{H}^m$; `alexandrov_entropy_degree_inequality` and `rcd_homological_index_entropy_inequality` specialize Part (1) to Alexandrov spaces and Mitsuishi's homological index.
2. **Equality-Case Rigidity (`ae_jacobian_eq_one_of_normalized_equality`, `rcd_entropy_degree_equality_rigidity`).** Combines the finite-cover degree reduction (`finite_cover_entropy_degree_reduction`), homothetic normalization (`homothetic_normalization`), the a.e. mass-Jacobian sandwich, degree surjectivity (`surjective_of_degree_ne_zero`), and Lipschitz-volume rigidity into a single theorem.

Listing 1: Master entropy-degree inequality and equality rigidity signatures in `RCDEntropyDegree.RCD`.

```
theorem rcd_entropy_degree_inequality
    (C : CMMPOrientationCurrent X K N CurrentX) [IsFiniteMeasure C.hausdorffMeasure]
    (FY : FundamentalManifoldCurrent Y CurrentY) {h_Xbar h_Ytilde : ℝ} (hh_Ytilde : 0 < h_Ytilde)
    {deg_f : ℤ} (F_s : ℝ → X → Y) (D_s : ∀ s : ℝ, s > h_Xbar → LipschitzDegreePushforward C FY (F_s s))
    (Jac_s : ℝ → X → ℝ) (h_deg_homotopy : ∀ (s : ℝ) (hs : s > h_Xbar), (D_s s hs).lipschitzDegree = deg_f)
    (h_mass_push : ∀ (s : ℝ) (hs : s > h_Xbar),
      FY.totalMass (D_s s hs).pushforwardCurrent ≤ ∫ x, Jac_s s x ∂C.hausdorffMeasure)
    (hJac_int : ∀ s > h_Xbar, Integrable (Jac_s s) C.hausdorffMeasure)
    (hJac_bound : ∀ s > h_Xbar, ∀ᵐ x ∂C.hausdorffMeasure, Jac_s s x ≤ (s / h_Ytilde) ^ N) :
    |(deg_f : ℝ)| * h_Ytilde ^ N * FY.manifoldVolume ≤ h_Xbar ^ N * C.totalVolume

theorem rcd_entropy_degree_equality_rigidity [CompactSpace X] [T2Space Yhat]
    (C_rescaled : CMMPOrientationCurrent X K N CurrentX) [IsFiniteMeasure C_rescaled.hausdorffMeasure]
    (FYhat : FundamentalManifoldCurrent Yhat CurrentYhat)
    {k deg_fhat h_Xbar h_Ytilde vol_X vol_Y : ℝ} (hk_pos : 0 < k) (hh_Xbar : 0 < h_Xbar) (hh_Ytilde : 0 < h_Ytilde)
    (hdeg_mult : k = k * deg_fhat) (hvol_cover : FYhat.manifoldVolume = k * vol_Y)
    (heq : entropyVolumeProduct N h_Xbar vol_X = k * entropyVolumeProduct N h_Ytilde vol_Y)
    (hvol_rescaled : C_rescaled.totalVolume = (h_Xbar / h_Ytilde) ^ N * vol_X)
    {Fhat_inf : X → Yhat} (hFhat_cont : Continuous Fhat_inf)
    (D_inf : LipschitzDegreePushforward C_rescaled FYhat Fhat_inf) (Jac_inf : X → ℝ)
    (hD_deg_ne : D_inf.lipschitzDegree ≠ 0) (hD_deg_abs : |(D_inf.lipschitzDegree : ℝ)| = deg_fhat)
    (h_open_pos : ∀ U : Set Yhat, IsOpen U → U.Nonempty → 0 < (FYhat.volumeMeasure U).toReal)
    (h_supp : (FYhat.massMeasure (D_inf.lipschitzDegree • FYhat.fundamentalClass) (Set.range Fhat_inf)ᶜ).toReal = 0)
    (h_mass_push : FYhat.totalMass D_inf.pushforwardCurrent ≤ ∫ x, Jac_inf x ∂C_rescaled.hausdorffMeasure)
    (hJac_int : Integrable Jac_inf C_rescaled.hausdorffMeasure)
    (hJac_le_one : ∀ᵐ x ∂C_rescaled.hausdorffMeasure, Jac_inf x ≤ 1)
    (h_lip_vol_rigidity : Function.Surjective Fhat_inf → C_rescaled.totalVolume = FYhat.manifoldVolume →
      (∀ᵐ x ∂C_rescaled.hausdorffMeasure, Jac_inf x = 1) →
      ∀ x₁ x₂ : X, dist (Fhat_inf x₁) (Fhat_inf x₂) = (h_Xbar / h_Ytilde) * dist x₁ x₂) :
    let c := h_Xbar / h_Ytilde; 0 < c ∧ deg_fhat = 1 ∧ c⁻¹ * h_Xbar = h_Ytilde ∧
    c ^ N * vol_X = FYhat.manifoldVolume ∧ C_rescaled.totalVolume = FYhat.manifoldVolume ∧
    (∀ᵐ x ∂C_rescaled.hausdorffMeasure, Jac_inf x = 1) ∧ Function.Surjective Fhat_inf ∧
    (∀ x₁ x₂ : X, dist (Fhat_inf x₁) (Fhat_inf x₂) = c * dist x₁ x₂)
```

3 Architectural Synthesis: Unconditional Proofs vs. Interfaces

1. Unconditionally Proved Pillars. All metric-measure volume entropy bounds, Bishop–Gromov hyperbolic growth estimates, normalized exponential weight π_1-equivariance, Sturm's CAT(0) d^2-barycenter existence/uniqueness (via the CAT(0) variance inequality `sturm_variance_inequality`), geodesic homotopy descent to

Architectural Chapter	Nodes	Principal Proved Achievements (#1CAC78)
Ch. 1: Volume Entropy & Scaling	4	Basepoint independence, hyperbolic Bishop–Gromov growth density bound, isometric/equivariant exponential weights, homothetic & finite-cover reduction (`MetricMeasure/VolumeEntropy.lean`).
Ch. 2: Sturm CAT(0) **Barycenter**	4	Renormalized d^2-functional continuity, linear growth, midpoint deficit & Sturm variance uniqueness, π_1-equivariant natural maps, geodesic homotopy descent (`MetricMeasure/SturmBarycenter.lean`).
Ch. 3: RCD(K,N) **& CMMP Currents**	4	Codimension-two singular stratification $\mathcal{S}(X) \subseteq \mathcal{S}^{N-2}$, Alexandrov specialization, and CMMP orientation current mass & singular-nullity laws (`RCD/StructureAndCurrents.lean`).
Ch. 4: Degree & Homotopy Invariance	4	Regular value openness & full measure ($V_f = Y \setminus f(\mathcal{S}(X))$), Lipschitz pushforward degree identity, Wenger homotopy invariance, surjectivity, and homological index (`RCD/DegreeTheory.lean`).
Ch. 5: Jacobians & Rigidity	5	Metric Jacobian, pushforward mass-Jacobian bound (Lemma 3.3), pointwise entropy-Jacobian bound via `BCG`, Theorem 1.1 Parts 1–2 (`RCD/EntropyDegreeRigidity.lean`).

Table 2: Breakdown of the 21 blueprint nodes (15 proved theorem nodes, 6 definition nodes).

quotients, regular-value openness/full-measure lemmas, algebraic pushforward degree identities, the finite-cover & homothetic rescaling reductions, the mass-Jacobian sandwich (`ae_jacobian_eq_one_of_normalized_equality`), and the upstream rank-one Busemann Hessian determinant bound dispatched via `BCG` are proved unconditionally from Mathlib first principles.

2. Typed Metric-Measure & Geometric-Measure Interfaces. Deep metric-geometric foundational theorems not yet available in Mathlib 4 are encapsulated inside explicit hypothesis structures without introducing any `axiom` declarations: `NonCollapsedRCDStructure` / `AlexandrovSpaceWithoutBoundary` (De Philippis–Gigli regularity and codimension-two stratification [4]), `CMMPOrientationCurrent` (the CMMP integral N-current on closed orientable RCD(K,N) spaces [5]), `LipschitzDegreePushforward` (Sormani–Wenger / Kirchheim area and homotopy constancy [3, 6]), and the explicit Lipschitz-volume rigidity hypothesis `h_lip_vol_rigidity` (Li–Wang / Connell–et al. [7, 6]) in Theorem 1.1 Part 2.

4 Modular Dependency Architecture and Verification Pipeline

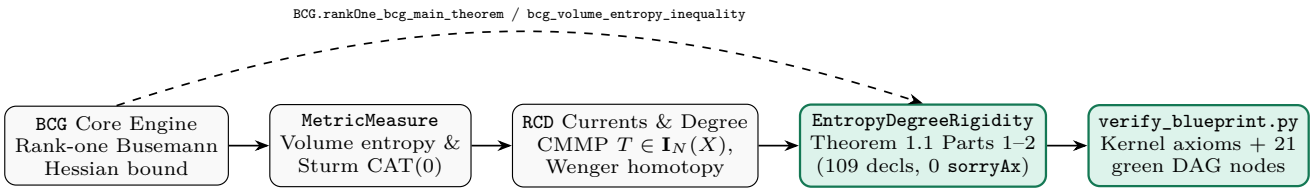


Figure 1: Package architecture linking `BCG` (`formalized-barycenter-methods`) to `RCDEntropyDegree` and automated verification.

Downstream applications of the Alexandrov entropy-degree theorem (Suárez-Serrato, `arXiv:2606.30403`) consume `RCDEntropyDegree` in the companion repository `alexandrov-entropy-degree-applications`.

References

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A Blueprint Dependency Graph

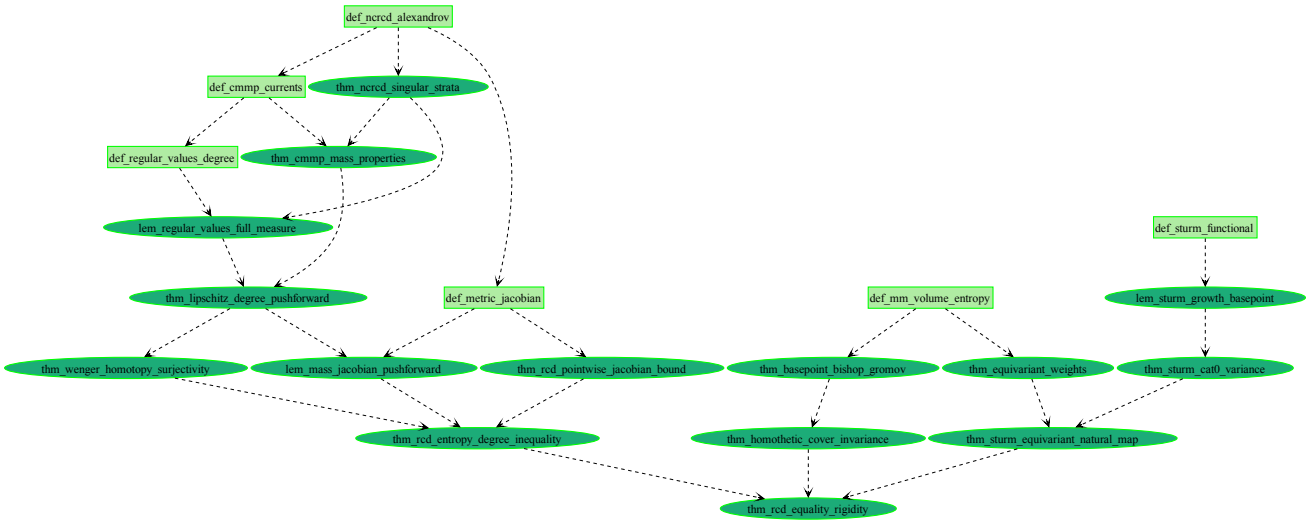


Figure A.1: Complete `leanblueprint` Dependency Graph (Portrait View). Generated from `blueprint/src/content.tex` via `build_dep_graph.py` (interactive HTML version at `blueprint/web/dep_graph_document.html`). Arrows point from each foundational definition or lemma to the downstream theorems that depend on it. **Node shapes and colors: Rectangles (6 nodes, green `#B0ECA3`):** Formalized mathematical definitions. **Ellipses (15 nodes, dark green `#1CAC78`):** Fully machine-checked theorems, lemmas, and certificates compiling with `0 sorryAx` and standard Lean 4 axioms across the entire dependency closure. **Unproved nodes:** 0 light green (`#9CEC8B`), 0 light blue (`#A3D6FF`).